

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>
the total size of the file is 200 GB
the download speed is 2 GB / minute
at 40 % of the way through , Carla has downloaded $0.4 * 200 = 80$ GB
the remaining size of the file is $200 - 80 = 120$ GB
since the download speed is 2 GB / minute , the time taken to download 120 GB is $120 / 2 = 60$ minutes
the time taken to download the first 80 GB is $80 / 2 = 40$ minutes
the total time taken is the sum of the time taken to download the first 80 GB and the time taken to download the remaining 120 GB , which is $40 + 60 = 100$ minutes
the time taken to install the windows updates is 20 minutes
the total time taken is $100 + 20 = 120$ minutes
</think>< answer >
120
</answer>

